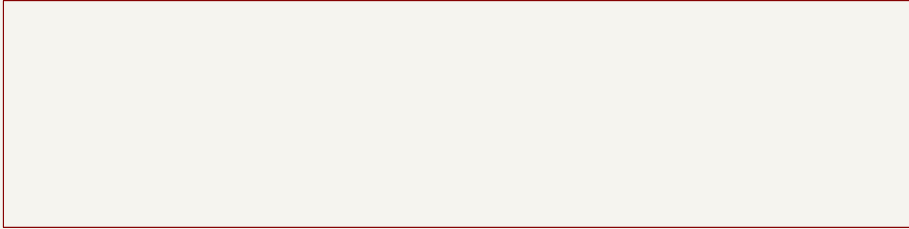


ESP32 EMU TURBO

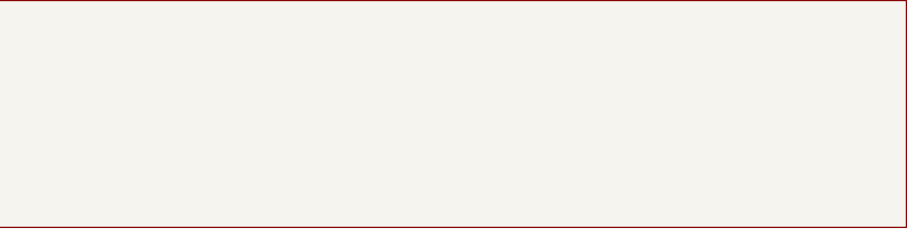
Hierarchical root schematic – all sub-sheets linked below

Power Supply



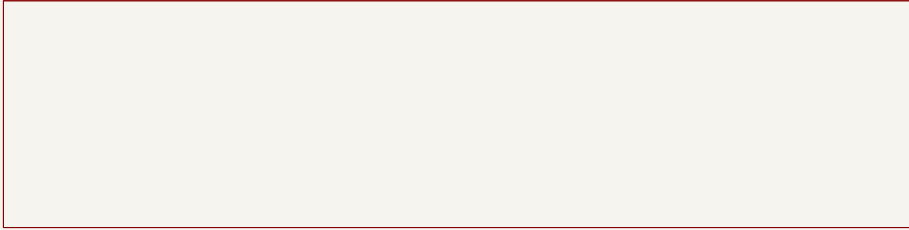
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Mcu



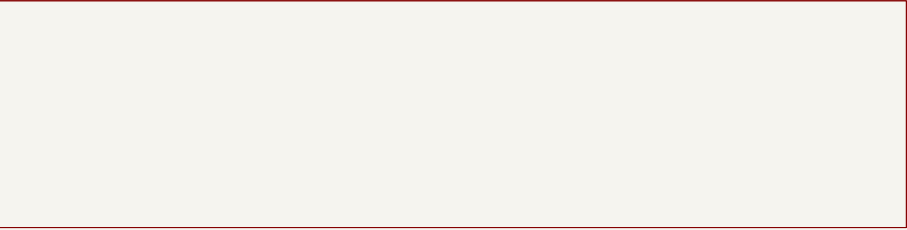
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Display



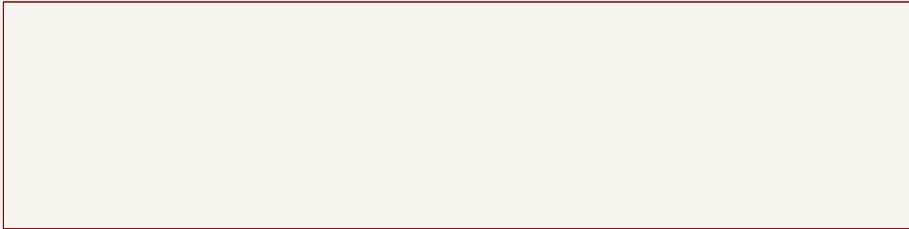
File: 03-display.kicad_sch

Audio



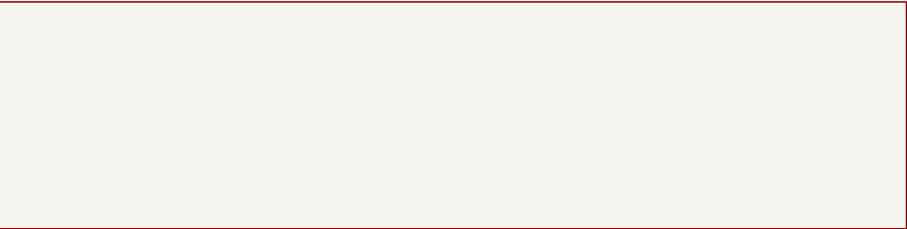
File: 04-audio.kicad_sch

Sd Card



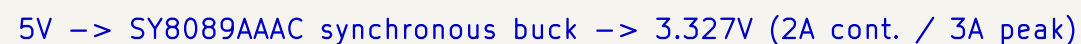
File: 05-sd-card.kicad_sch

Controls



File: 06-controls.kicad_sch

USB-C -> IP5306 SOP-8 (charge + boost) -> SY8089AAAC buck -> 3.3V rail

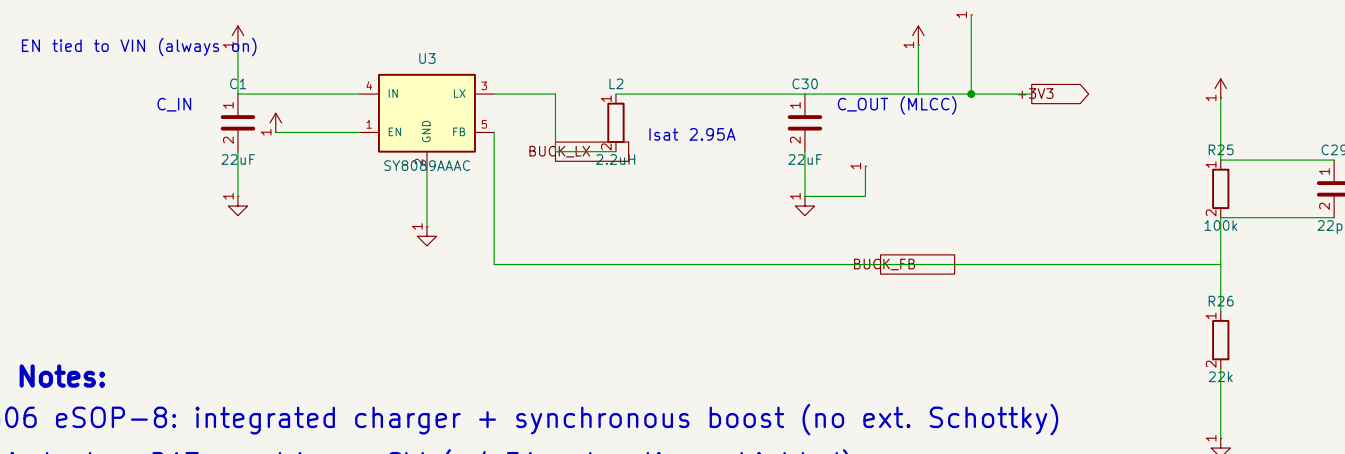


Design Notes:

- ## FEEDBACK DIVIDER

$$V_{out} = 0.6 \times (1 + 100k/22k) = 3.327V$$

C29 = datasheet feed-forward cap (load transient)



Generated by scripts/generate_schematics
ESP32 Emu Turbo – Handheld Retro Console

Sheet: /Power Supply/
File: 01-power-supply.kicad_sch

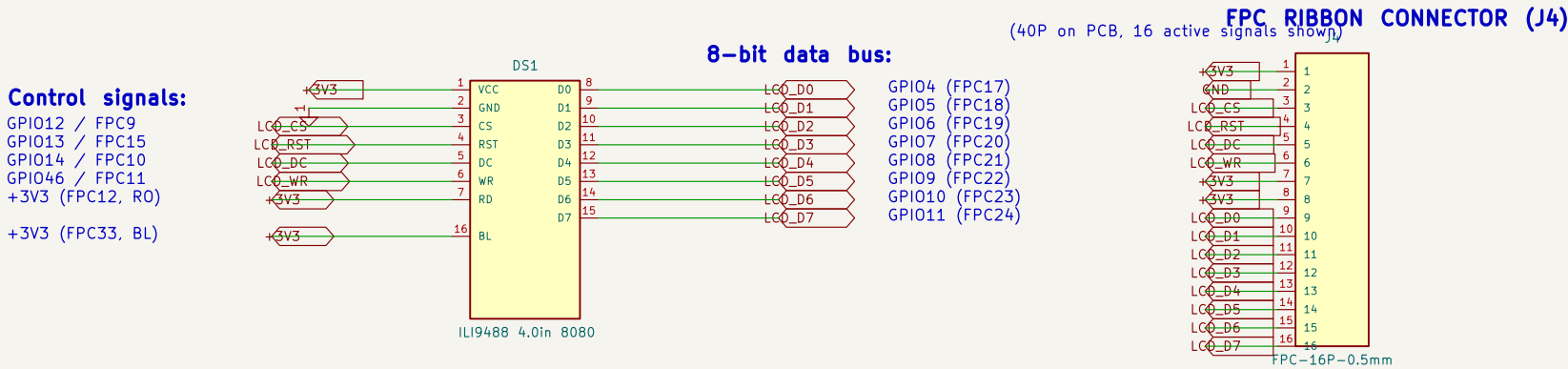
Title: Power Supply

Size: A3	Date:
KiCad E.D.A. 9.99.0-5542-g9c1654c496	

Rev:
Id: 2/7

DISPLAY – ILI9488 4.0in 320x480

8-bit 8080 parallel interface (mandatory for SNES emulation speed)



ILI9488 Panel FPC-40P Pin Mapping (source: components.md):

Panel 6,7=VDDI/VDDA(+3V3) 9=CS 10=DC 11=WR 12=RD 15=RESET

Panel 17-24=DB0-DB7 33=LED-A(BL) 34-36=LED-K(GND) 38,39=IM0,IM1(+3V3) 40=IM2(GND)

Interface mode: IM2=0 IM1=1 IM0=1 = 8080 8-bit parallel

- GPI04-11 form contiguous 8-bit bus for efficient register-level DMA
- WR strobes data on rising edge, RD tied HIGH (+3V3, write-only mode)

□ FPC PIN REVERSAL on PCB: panel pin N ↔ J4 pad (41-N).

e.g. panel pin 17 (DB0) contacts J4 pad 24. See datasheet_specs.py for the connector-side mapping.

Generated by scripts/generate_schematics
ESP32 Emu Turbo – Handheld Retro Console

Sheet: /Display/
File: 03-display.kicad_sch

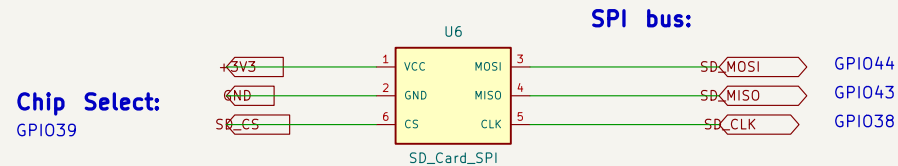
Title: Display – ILI9488 4.0in 8080 Parallel

Size: A3 Date:
KiCad E.D.A. 9.99.0-5542-g9c1654c496

Rev:
Id: 3/7

STORAGE — Micro SD Card

SPI interface for ROM storage (SNES ROMs up to 6MB)



Design Notes:

- SPI bus @ 20MHz (R2-MED-5: trace length ~150mm + 6 vias → 40MHz unreliable)
- SNES ROMs: 256KB to 6MB typical (HiROM/LoROM)
- FAT32 filesystem for easy ROM management via PC
- 3.3V logic: no level shifter needed (module has built-in)
- SPI pins: GPIO38/39 (CLK/CS) + GPIO43/44 (MISO/MOSI) – split across WROOM-1 Left and right
- y2 TODO (R4-MED-1): add local 100nF bypass within 5 mm of U6 VCC

Generated by scripts/generate_schematics
ESP32 Emu Turbo — Handheld Retro Console

Sheet: /Sd Card/
File: 05-sd-card.kicad_sch

Title: Storage — SD Card SPI
Size: A4 Date:

KiCad E.D.A. 9.99.0-5542-g9c1654c496

Rev:
Id: 4/7

MCU – ESP32–S3–WROOM–1 N16R8

Dual–core LX7 @ 240MHz, 16MB Flash, 8MB PSRAM

GPIO ASSIGNMENT TABLE

Display (8080 parallel):
GPIO4–11 = D0–D7 (8–bit data bus)
GPIO12=CS GPIO13=RST GPIO14=DC
GPIO46=WR (RD/BL tied +3V3 on FPC)

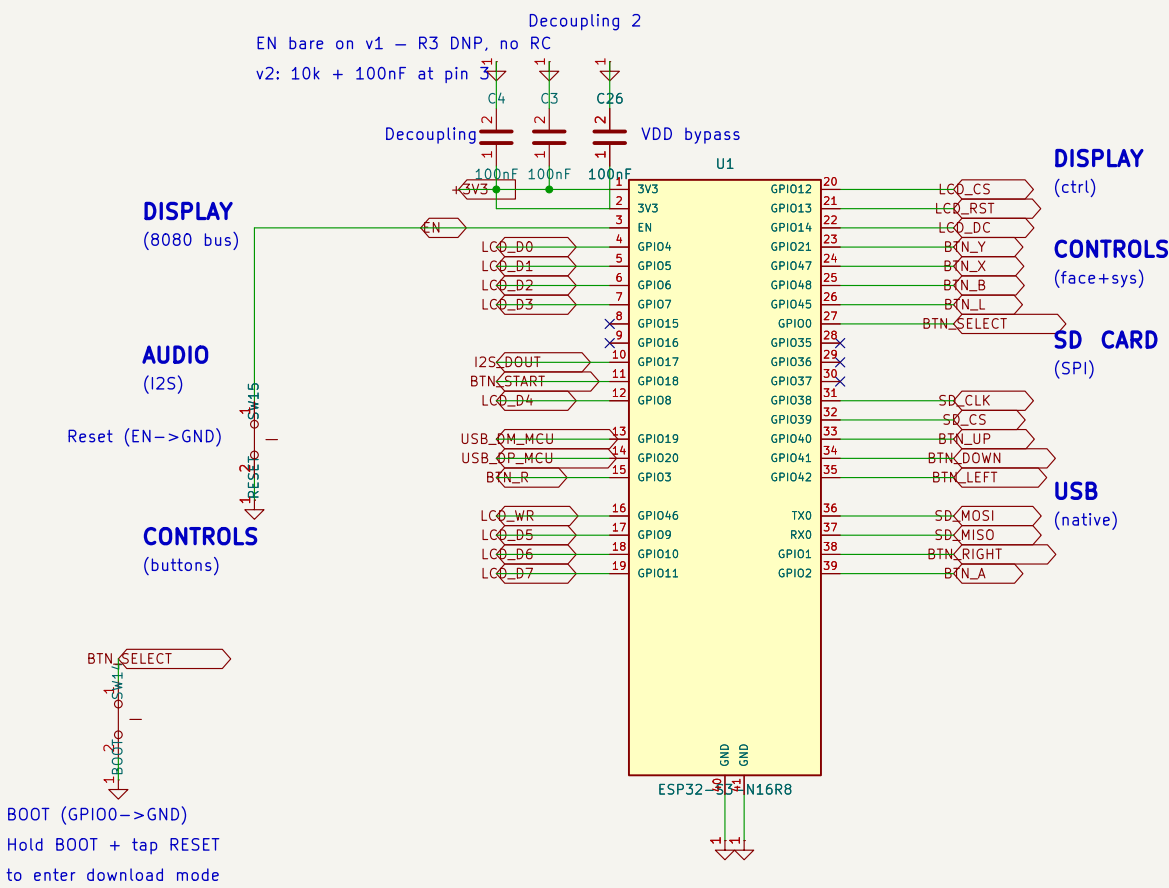
Audio (I2S):
GPIO15=BCLK GPIO16=LRCK GPIO17=DOUT

SD Card (SPI):
GPIO44=MOSI GPIO43=MISO GPIO38=CLK GPIO39=CS

Controls (active low, 10k pull–up):
D–pad: GPIO40=UP 41=DOWN 42=LEFT 1=RIGHT
Face: GPIO2=A 48=B 47=X 21=Y
Sys: GPIO18=START 0=SELECT 45=L 3=R

USB (native, firmware flash + debug):
GPIO19=D– GPIO20=D+

Reserved (do not use, module–internal):
GPIO26–32 = Octal Flash (N16)
GPIO33–37 = Octal PSRAM (R8)



Generated by scripts/generate_schematics
ESP32 Emu Turbo – Handheld Retro Console

Sheet: /Mcu/
File: 02–mcu.kicad_sch

Title: MCU – ESP32–S3–WROOM–1 N16R8

Size: A3 Date: KICad E.D.A. 9.99.0–5542–g9c1654c496

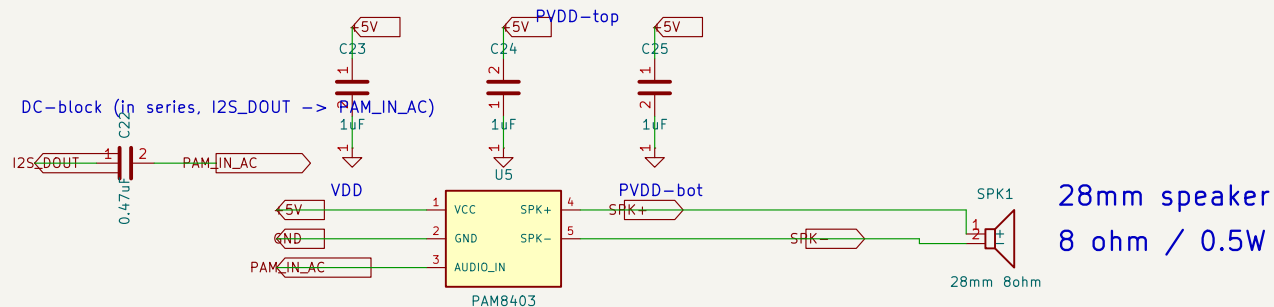
Rev: Id: 5/7

AUDIO OUTPUT

I2S DAC -> PAM8403 Class-D Amplifier -> 28mm 8ohm Speaker

I2S Bus (PDM TX mode - DOUT only):

I2S_BCLK GPIO15 - Bit Clock (UNUSED - PDM mode, v2 free)
I2S_LRCK GPIO16 - L/R Word Select (UNUSED - PDM mode, v2 free)
I2S_DOUT GPIO17 - PDM sigma-delta out @ 32kHz -> C22 -> PAM_IN_AC



PAM8403 Passives:



R20/R21 bias the amplifier inputs to PAM_VREF (R4-HIGH-3).

Design Notes:

- PAM8403: filterless Class-D stereo amplifier (3W per channel max)
- Only one channel used for mono audio
- No external DAC needed: ESP32-S3 has built-in I2S with DMA
- Powered from +5V rail for max headroom (3.3V limits volume)
- DMA-driven audio streaming for low CPU overhead

Generated by scripts/generate_schematics
ESP32 Emu Turbo - Handheld Retro Console

Sheet: /Audio/
File: 04-audio.kicad_sch

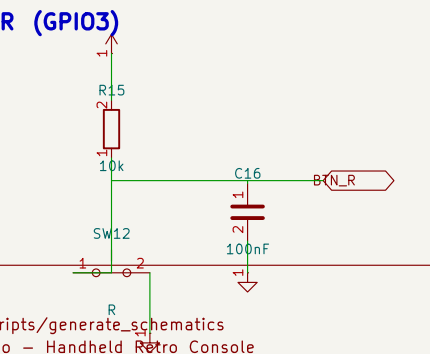
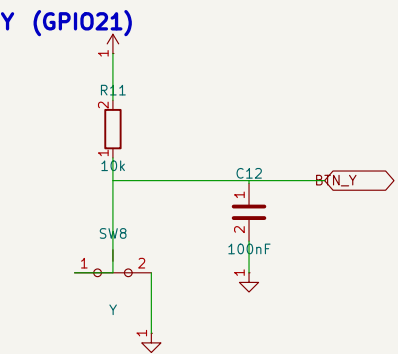
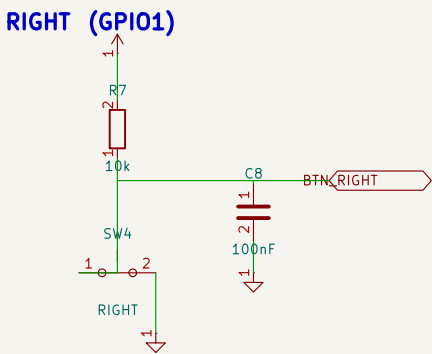
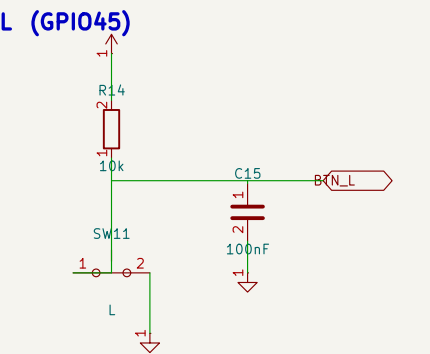
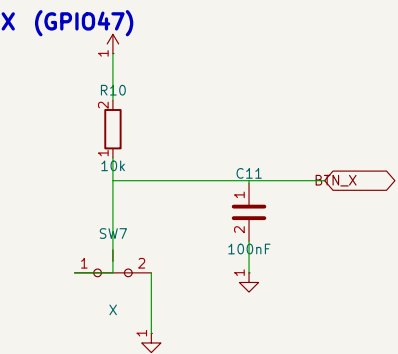
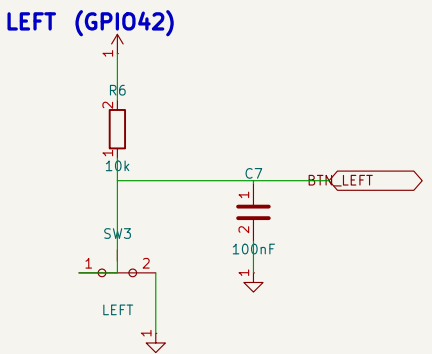
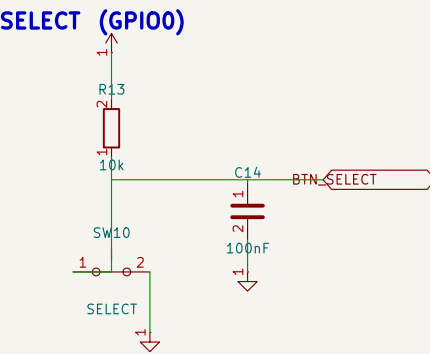
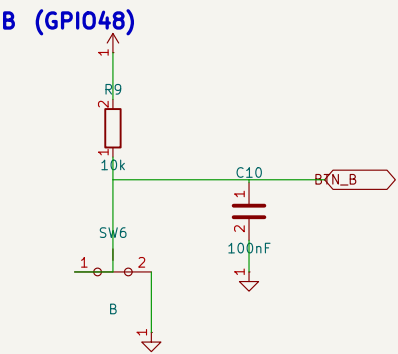
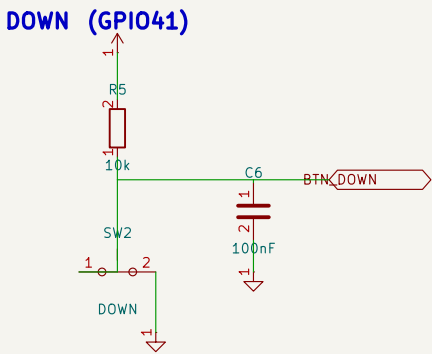
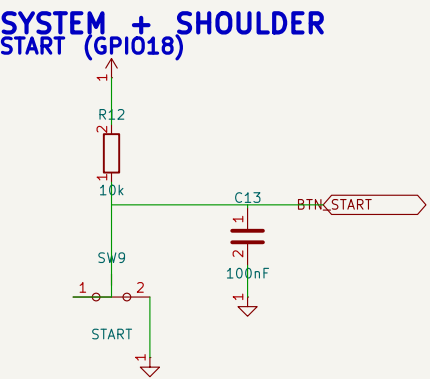
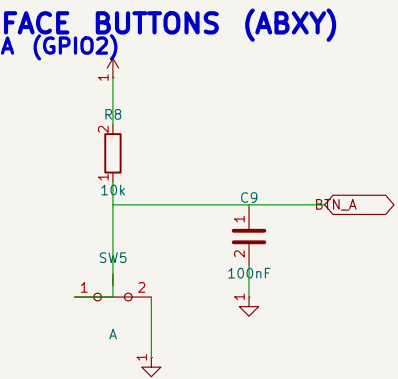
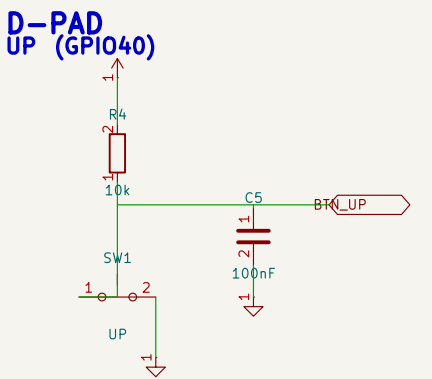
Title: Audio - I2S -> PAM8403 -> Speaker

Size: A4 Date:
KiCad E.D.A. 9.99.0-5542-g9c1654c496

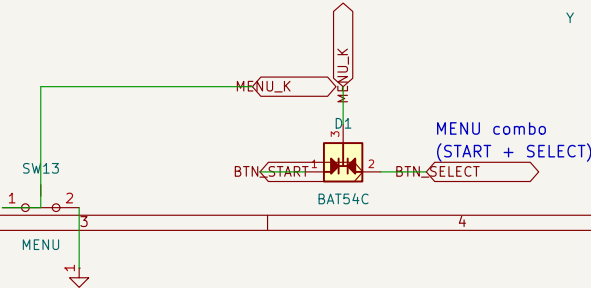
Rev:
Id: 6/7

CONTROLS – 12 Tact Switches

Active-low with 10k pull-up + 100nF debounce per button



MENU BUTTON



Generated by scripts/generate_schematics
ESP32 Emu Turbo – Handheld Retro Console

Sheet: /Controls/
File: 06-controls.kicad_sch

Title: Controls – 12 Buttons (SNES Layout)

Size: A3 | Date: | KiCad E.D.A. 9.99.0-5542-g9c1654c496

Rev: | Id: 7/7

BUTTON CIRCUIT (repeated 12x):